

EXPERIENCES

- Data Scientist, TESCO | Top 5 Performer (Top 2.5%) among 200+ Data Scientists** Winter 2024 – Present
- **Evaluation & Alignment:** Built an **LLM-as-a-Judge evaluation framework** to measure safety and quality, rigorously calibrating model scoring against human judgement to ensure alignment with business values
 - **Data-Centric AI:** Designed high-signal data collection workflows to construct a recipe-to-product knowledge graph; transformed **messy, unstructured** recipe content into structured data for RAG systems
 - **Model Improvement:** Diagnosed data distribution shifts in recommendation pipelines; designed targeted data augmentation strategies that solved cold-start problems, boosting NDCG by **0.73%** and coverage by **32.51%**
 - **Impact:** Delivered PySpark ETL pipelines generating **£5M+ annualized savings**, demonstrating ability to ship scalable, profitable ML solutions
- Off-Cycle Intern, Global Banking & Markets Department of Goldman Sachs** Autumn 2024
- Developed automated market-intelligence pipelines using Slang, converting unstructured financial streams into structured daily insights to replace manual email-based workflows

PROJECTS

- SETA: Scaling Environments for Terminal Agents, CAMEL-AI.org** Winter 2025
- **SOTA Benchmarking:** Contributed to the development of a terminal agent harness that achieved **#1 on Terminal-Bench 2.0** (46.5% success with Claude 4.5) and **Terminal-Bench 1.0** (35.0% with GPT-4.1)
 - **Self-Evolving Environment Synthesis:** Developed a scalable self-play pipeline to dynamically co-evolve task complexity with agent capabilities, maintaining training signals within the targeted reward zone to provide an adaptive curriculum for better reinforcement learning training performance
 - **Resilience & Failure Diagnosis:** Systematically diagnosed critical agent failure modes, including unproductive reasoning loops and OOD knowledge gaps; mitigated these through the implementation of state-management primitives and browser-augmented search tools
 - **RL-Training:** Collaborated with researchers to develop the RL pipeline, including **distributed PPO/GRPO training, FSDP-based model sharding, rollout formats, reward design, and Docker-isolated execution**
- Large Language Models for Product Substitution, UCL & TESCO** Summer 2024
- Transformed product substitution into a solvable NLP challenge, using LLMs (GPT-4o and Llama 3) with a **RAG-based, augment-then-summarise approach** to inject model knowledge into product metadata
 - Prepared and processed a historical user behavior dataset to **fine-tune a cross-encoder model** (ms-marco-MiniLM-L6-v2) with binary cross entropy loss, improving the re-ranking of candidate substitutions
 - Demonstrated significant performance gains over baselines in experiments, achieving **+0.73% NDCG and +32.51% Coverage**, validating the effectiveness of **productionising research** for a real-world business problem
- Blockchain Coder: Enhancing Code Generation with Multi-Agent Framework, UCL** Summer 2024
- **Designed a multi-agent framework** (Planner, Optimizer, Debugger) to improve true generalisation over simple memorization, addressing the ~40% accuracy drop that LLMs exhibit on previously unseen coding problems
 - Deconstructed complex problems and refined code via unit-test-driven and question-based iteration, **doubling the pass@1 success rate** (to 6.90%) and increasing the private test case pass rate by **5.7x** (to 38.08%)

EDUCATION

- University College London | MSc Machine Learning |First Class** 2023 – 2024
- Multi-agent AI (85.5%), Information Retrieval and Data Mining (87.5%), Bayesian Deep Learning (83.5%)
- Imperial College London | BSc Mathematics | First Class (80.3%)** 2020 – 2023
- Methods for Data Science (94.4%), Mathematical Foundations of Machine Learning (90%)

TECHNICAL SKILLS

Languages	Python, SQL, R, Julia, Mandarin (Native), Cantonese (Native)
Technologies	PyTorch, TensorFlow, HuggingFace, LangChain, AutoGen, vLLM, Milvus, Langfuse, PySpark, Pandas, Airflow, Docker, Git, RAG, Fine-tuning (SFT, PEFT, DPO)

ACHIEVEMENTS

- **ElevenLabs × LFH × Bolt Hackathon** ranked 2/50, created Sign2Speak an AI sign language translation system
- **Tech:Europe Hackathon** ranked 1/12 @ SalesAPE track, built SquareDeal, a AI-native marketplace app for SMBs
- **AI Content Creator** MarketVision, an open-source AI stock tracker, gained 1K+ followers through one tutorial